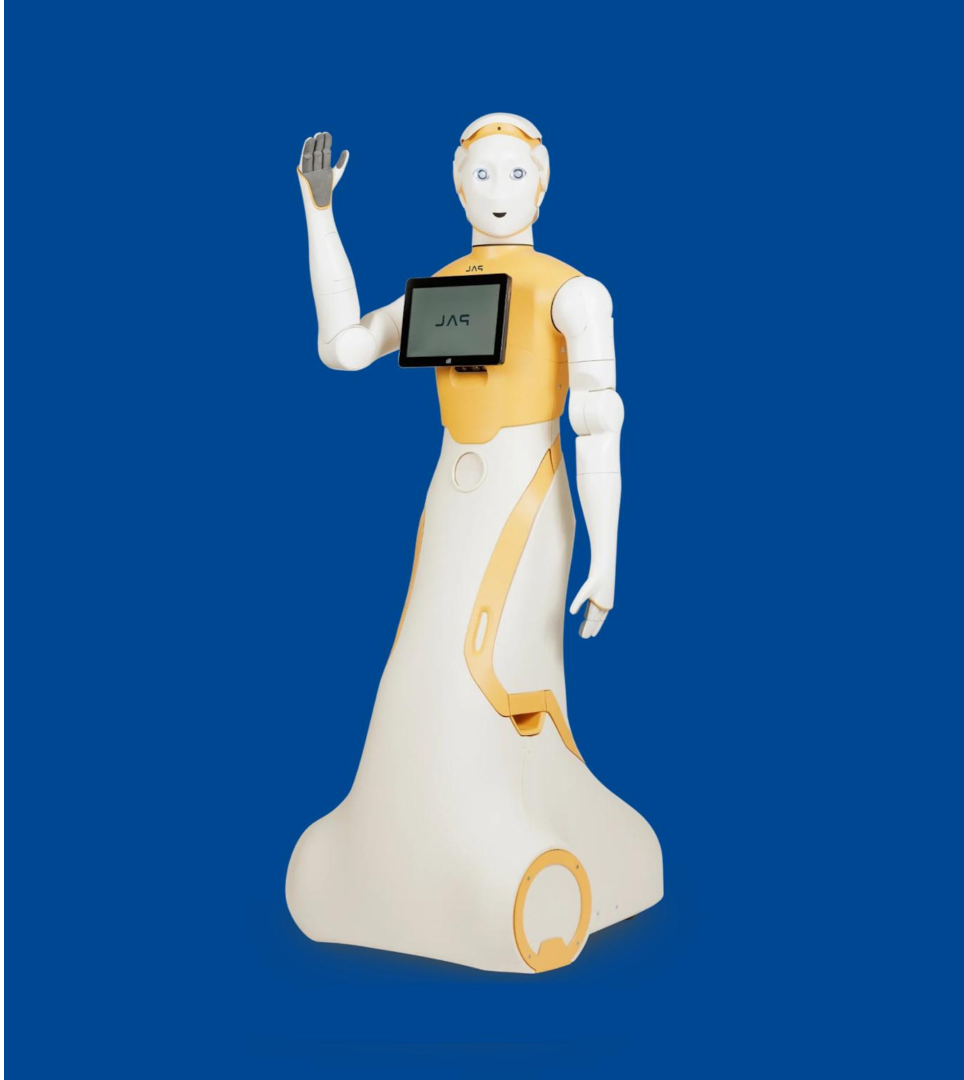
KTH VETENSKAP OCH KONST

ARI office assistant - Demo

DD2414 Engineering project in Robotics, Perception and Learning

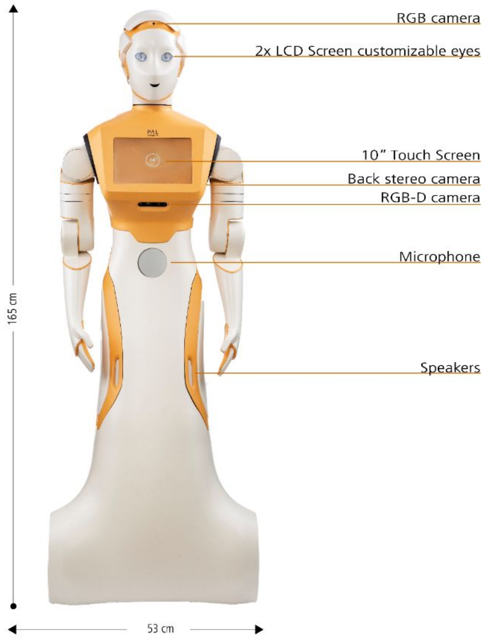
Content

- ARI from PAL robotics
- Project Goal and Milestones
- Progress
  - Behaviour Tree
  - Communication
  - Navigation
  - Human-Robot interaction
- Challenges
- Demo!



ARI Software and Hardware

ARI hardware



ARI software

Packages available in ROS noetic

- Navigation:
  - move_base
  - Map manager creates and uses 2d occupancy grids
  - Point clouds used for local obstacle avoidance

- Human-robot interaction: pyHRI
  - Multiface detection but single body detection (due to ROS1)
  - No voice recognition
  - Speech synthesis and speech recognition

Project goal and milestones

Project Goal

Implementing the functionality for ARI to act as an office assistant.

Ari can understand a speaker's intent, and execute the following commands:

- "Follow me"
- "Translate our conversation"
- "Show me where person X is"
- "Take me to office Y"
- "Explore the environment"
- "Go to …"

Ari looks at the speaker when communicating with them, and has a behaviour tree of tasks such that they can be interrupted, and Ari can recognize when help is needed for a task.

Milestones

Implementing the functionality for ARI to act as an office assistant.

Milestone 1: Text to speech, speech to text, basic mapping and navigation

Milestone 2: LLM for communication, look at speaker, remember faces

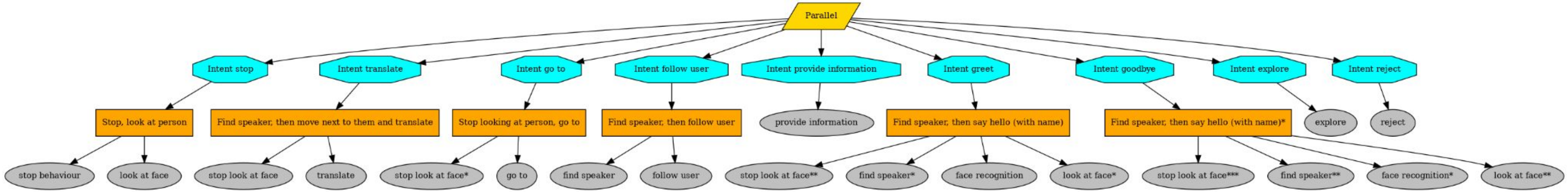
Milestone 3: Remember people's locations, follow a person, turn to speaker

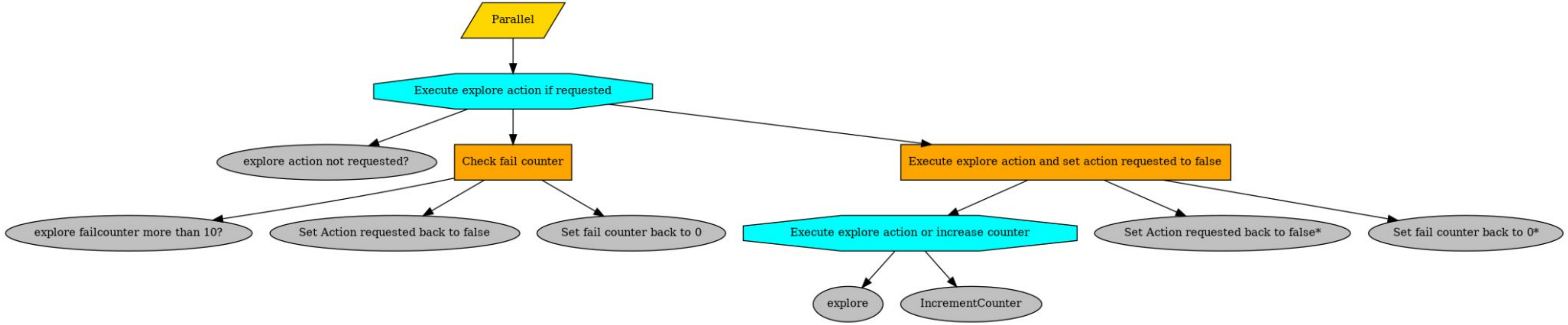
Milestone 4: Execute intents based on commands

Milestone 5: Tasks can be interrupted based on hierarchy

Current State

Behaviour Tree





Communication (LLM/intents)

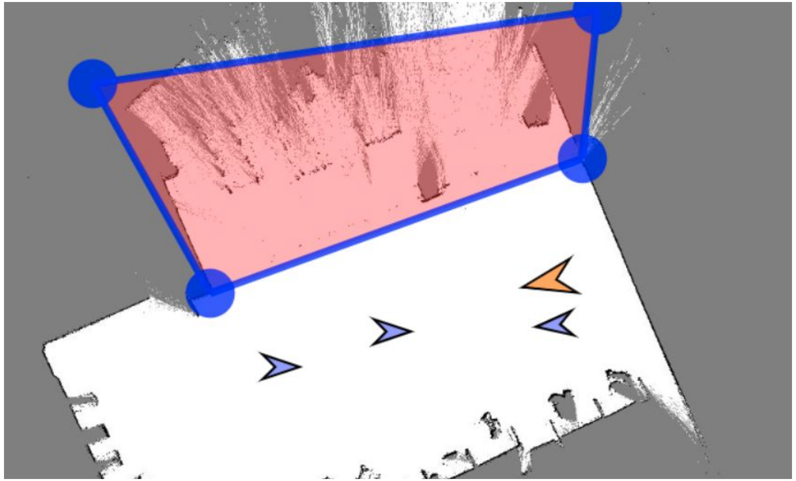
- Model: Mistral 7b
- Ollama running locally
- Intents:
  - greet
  - goodbye
  - remember user
  - provide information
  - go to
  - follow user
  - explore
  - find object
  - translate
- The LLM node publish a topic that contains the intent, phrase, language and the respective input (if necessary).

Mapping

- The use of the lidar and the RGBd onboard cameras to create a 2D map of the environment.

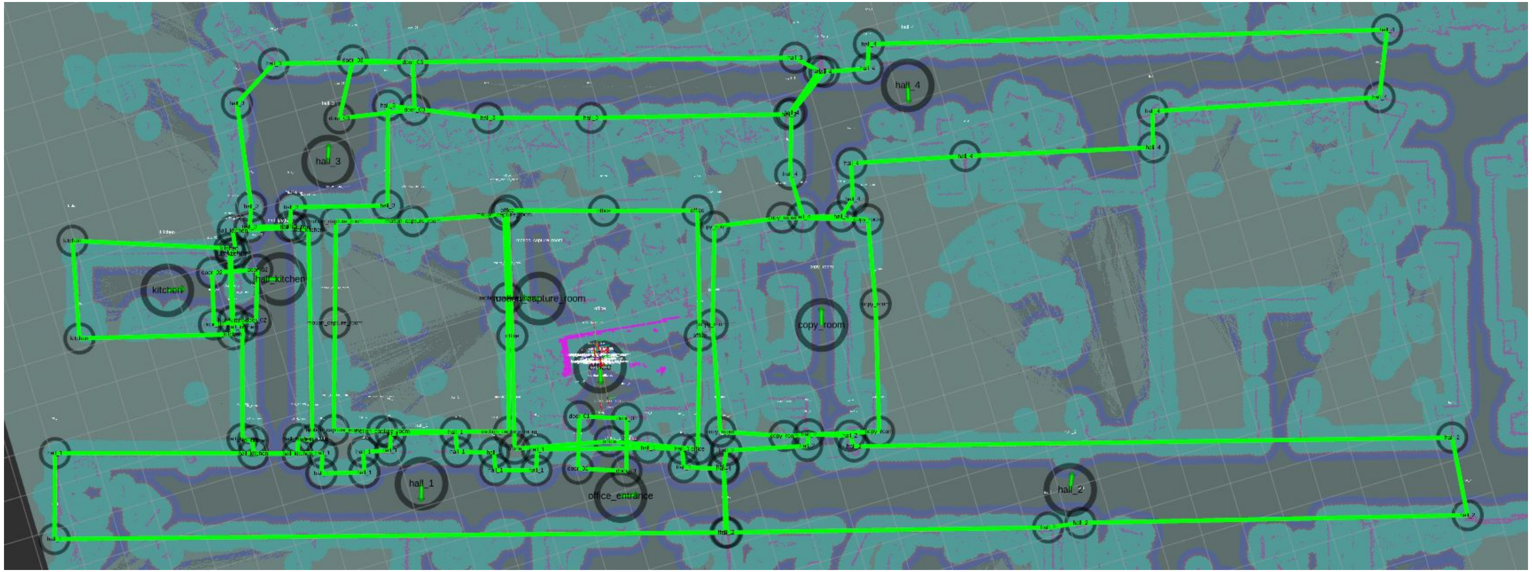
Navigation

- It is done to Point Of Interest (POI) previously defined in the map that was created.



Current Map

The map used is from the 4th floor of the RPL building. Several Zones of Interest and Points of Interest have been defined to assist with room navigation. Zones in doorways have been defined to adapt to different behaviour when crossing doors.



Speech-to-Text

Whisper, an open-source speech recognition system developed by OpenAI. It provides high-quality transcription across multiple languages and accents.

Whisper runs locally and does not require an internet connection, making it suitable for secure or offline environments.

Convert array of bytes containing the raw audio data captured by ARI’s microphones to text.

Text-to-Speech

Edge Text-to-Speech (edge_tts). When the language the speaker is using is recognized as English by the STT, it will use ARI’s incorporated TTS; but when the language is another (in this project we have enabled Swedish, German, Spanish, Japanese and French), then it will use Edge’s TTS.

Human-Robot interaction

- **Face Recognition:** Save face encodings to database and assign name (if known) and last seen location
- **Multibody recognition:** create a skeleton of all the bodies in view (YOLO n5)
- Look at person's face
- Follow person around
- Turn to speaker
  - Use direction of sound together with speech recognition node
  - Recognizes when body is facing ARI
  - Turns towards direction until it finds a person whose body is facing ARI
- Translation

Stop command

Stop behaviour was set up as highest priority on the tree.

Activation command

To not react to background noise or conversations, ARI will start listening after hearing the word robot in the first or second position of a sentence, e.g. “Hey robot!”

Challenges

Challenges

- Internal issues on ARI (screen,expressive eyes, low battery)
- Blurry images in face recognition
- Microphone positioning
- Language support
- Obstacle navigation
- Unfiltered voice input
- Planning
- Network
- Documentation
- Dynamic Environment

Results and Conclusions

Results

- Successfully demonstrated the concept of ARI as a mobile office assistant and translator.
- The robot can follow people, interpret conversational instructions, and maintain orientation toward the speaker.
- Language translation.
- People's location recording and navigation to the room they were last seen.
- Navigation to a specific room set on the map.